

ASD Longitudinal RMD

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2025-04-17

This Markdown summarizes information regarding the data that will be used for a longitudinal analysis of vocabulary acquisition in autistic children. Here we are interested in vocabulary sizes between 3 & 600 words produced. The numbers reported here differ from our first look because of a couple of reasons:

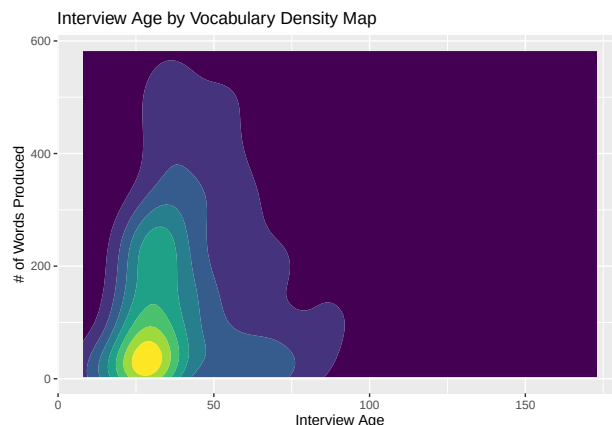
1. I recycled old code from a different project to combine the old and new GUIDs. This code had a part where we were only accepting the “best” CDI record (that with the highest number of words produced across multiple records). This was reducing the number of possible longitudinal data points. Good news, we have more kids with longitudinal data 9 - 15 months between CDIs.
2. I used an “|” instead of “&” when defining the 2 - 4 window for kids with more granular longitudinal data. This made it seem like we had much more kids with fine-tuned longitudinal data. After fixing this, we only have 6 kids with data at that level of specificity.

Demographic Information

Demographics for kids with between 3 & 600 productive vocabulary. The majority of kids have early interviews with relatively low productive vocabularies.

```
## [1] "Number of unique subjectkeys is 350 and the number of unique CDI entries is 629"
```

```
##      subjectkey      interview_age      nProduced
## Length:410116      Min.   :  8.00      Min.   :  3.0
## Class :character    1st Qu.: 26.00      1st Qu.: 31.0
## Mode  :character    Median : 36.00      Median :151.0
##                               Mean  : 43.23      Mean  :184.4
##                               3rd Qu.: 54.00      3rd Qu.:296.0
##                               Max.   :173.00      Max.   :582.0
```

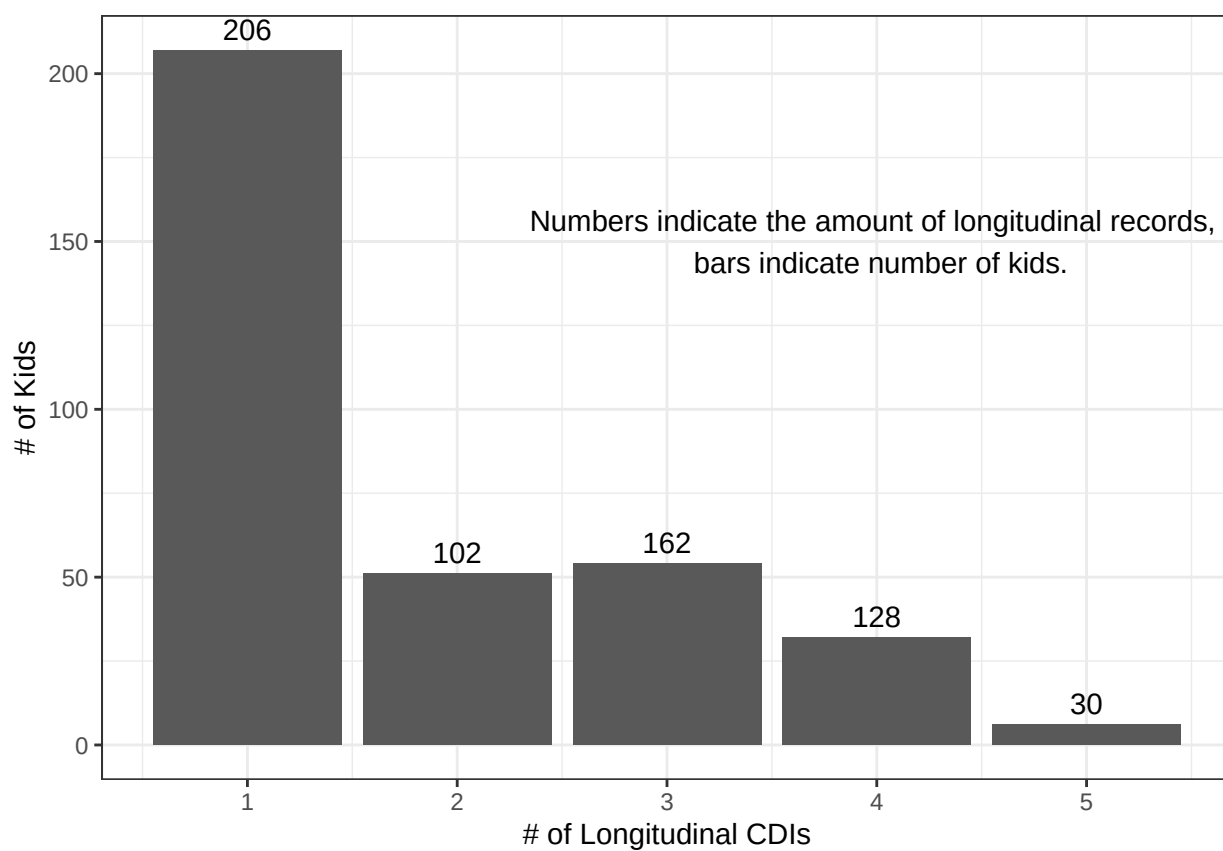


Longitudinal Analysis

Of the 628 CDIs, 206 only have an entry for a single age, removing them from longitudinal consideration. That leaves 422(111 unique kids) with at least 2 CDIs.

Table 1: CDI Longitudinal Counts

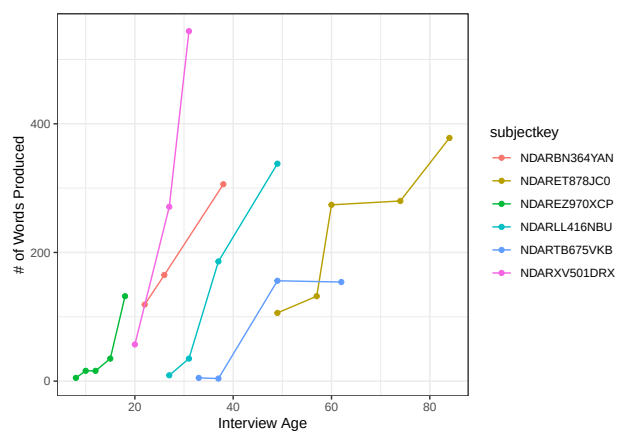
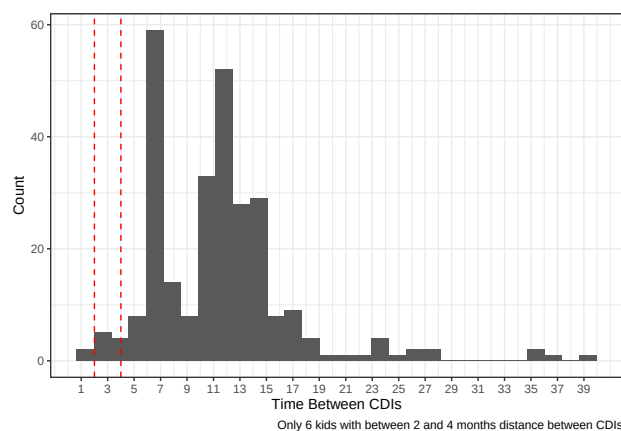
# of CDIs	# of Records	# of Kids
1	207	207
2	102	51
3	162	54
4	128	32
5	30	6



Kids with longitudinal data in 3 month intervals

There are only 6 kids with between 2 and 4 months between CDIs. This is different from our first look at this because we mistakenly used “|” in our filter instead of “&” when defining our time window. There are 9 total records with between 2 and 4 months in between CDIs, with the majority of those (4) being 4 months apart.

subjectkey	interview_age	diff	nProduced
NDARBN364YAN	22	NA	119
NDARBN364YAN	26	4	165
NDARBN364YAN	38	12	306
NDARET878JC0	49	NA	106
NDARET878JC0	57	8	132
NDARET878JC0	60	3	274
NDARET878JC0	74	14	280
NDARET878JC0	84	10	378
NDAREZ970XCP	8	NA	5
NDAREZ970XCP	10	2	16
NDAREZ970XCP	12	2	16
NDAREZ970XCP	15	3	35
NDAREZ970XCP	18	3	132
NDARLL416NBU	27	NA	9
NDARLL416NBU	31	4	35
NDARLL416NBU	37	6	186
NDARLL416NBU	49	12	338
NDARTB675VKB	33	NA	5
NDARTB675VKB	37	4	4
NDARTB675VKB	49	12	156
NDARTB675VKB	62	13	154
NDARXV501DRX	20	NA	57
NDARXV501DRX	27	7	271
NDARXV501DRX	31	4	544



Kids with longitudinal data in 12 month intervals

There are 111 kids with CDIs at the specificity of 9 - 15 months between CDIs. Below are tables and figures showing the distribution of time between CDIs both across participants and how many individual participants have multiple CDIs within the 9 - 15 time range. The majority of CDIs that meet the 9 - 15 month criteria fall in the 12-month time between CDIs. Additionally, the majority (79 of 111) only have one instance of records that are 9 - 15 months apart.

```
## [1] "There are 111 unique kids with CDIs 9 - 15 months between CDIs"
```

```
## [1] "There are 150 cases of CDIs being between 9 and 15 months apart. The counts are shown below."
```

Table 3: CDI Latencies Counts

Month Latency	Count
9	8
10	14
11	19
12	52
13	28
14	15
15	14

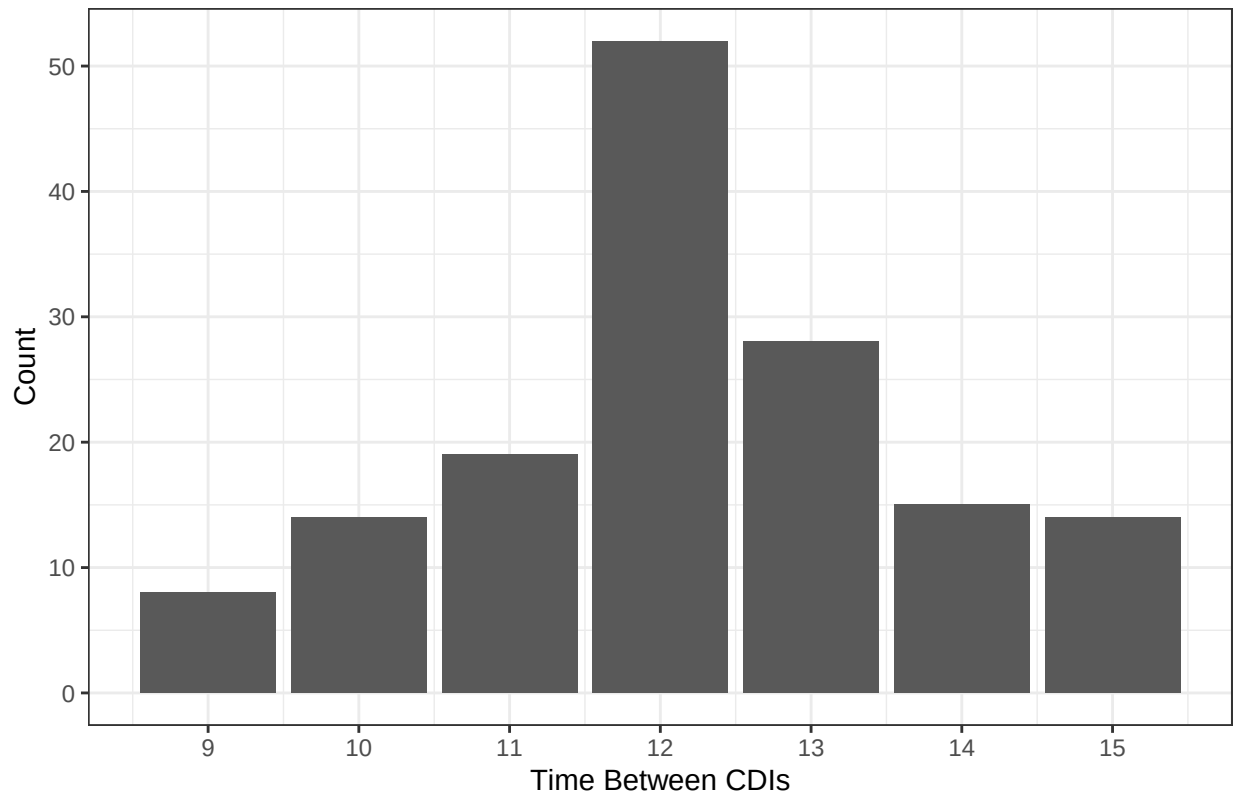
Table 4: Multiple Contributions Counts: Note: 1 contribution = 2 CDIs 9-15 months apart

# of Contributions	Count
1	79
2	25
3	7

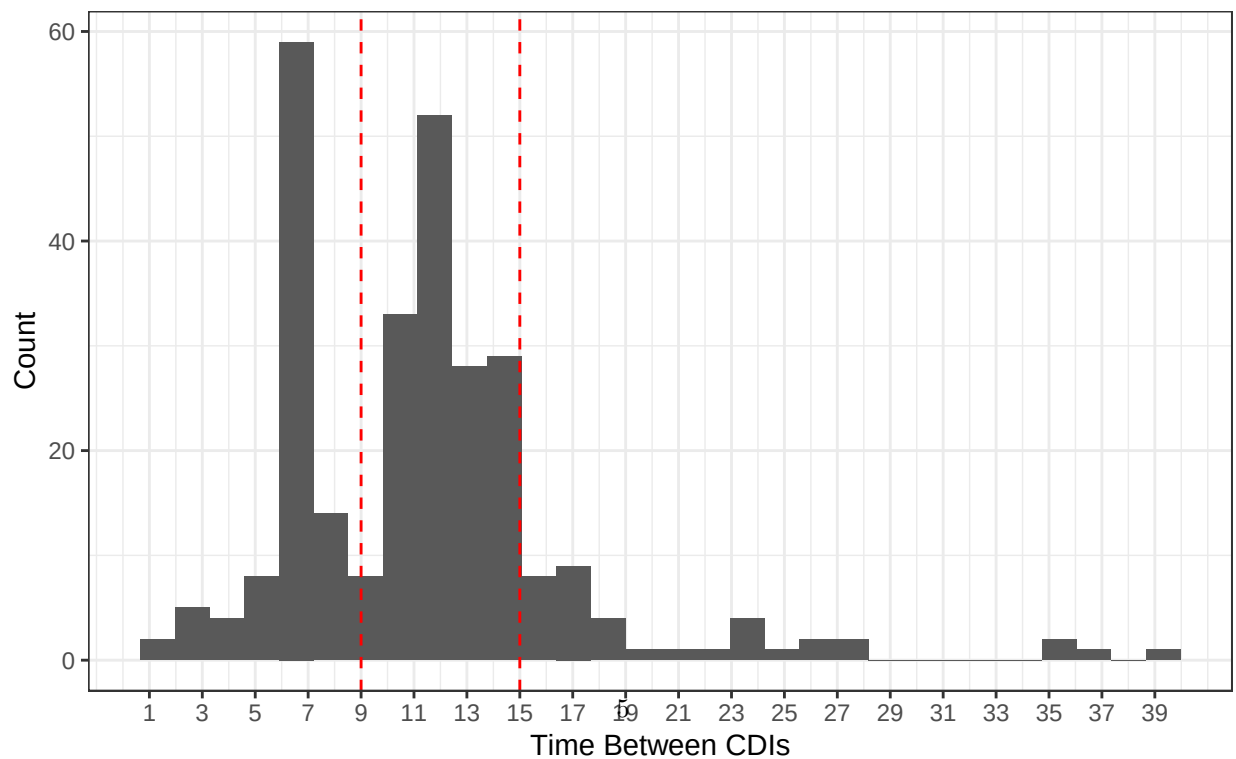
Histograms

Graphical representation of tables above. Plot on the bottom: red lines reflect the 9 - 15 month cutoffs.

Histogram of Times Between CDIs (9-15)

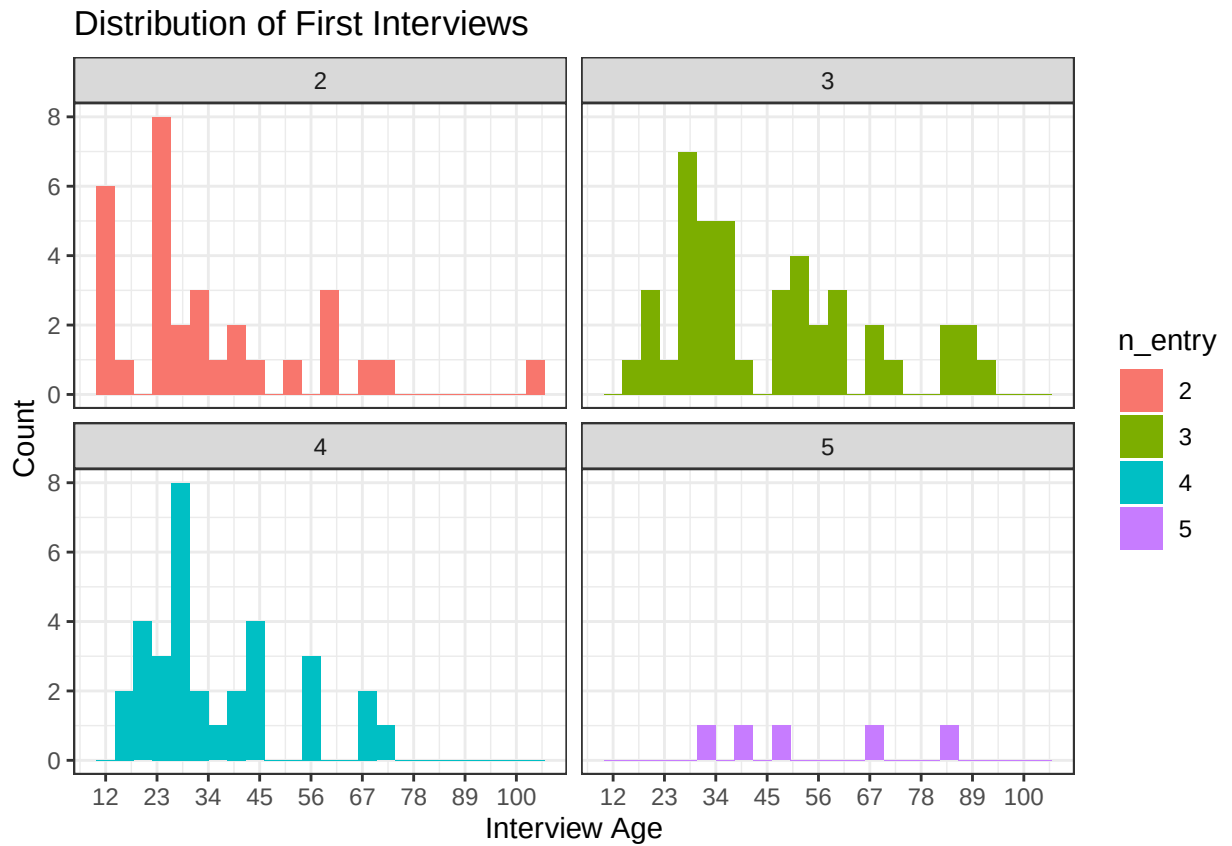
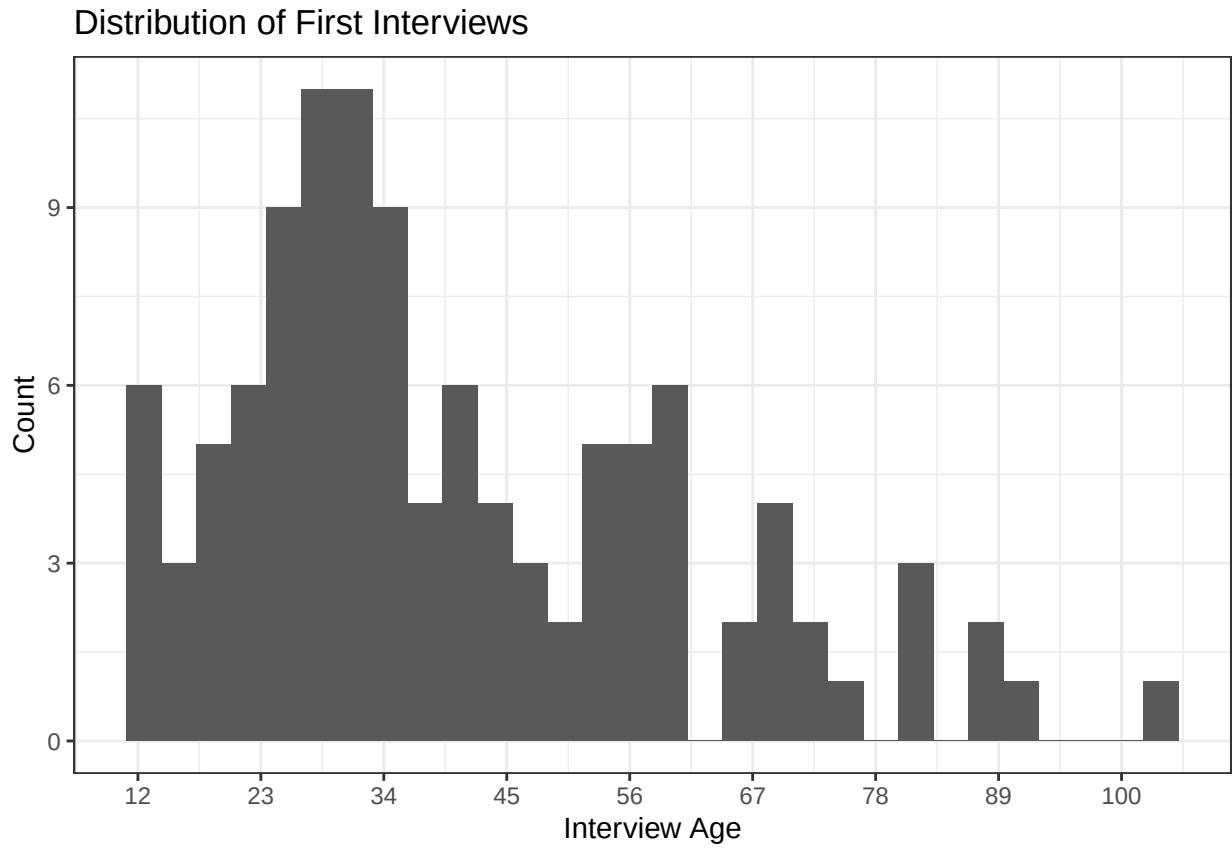


Histogram of Times Between CDIs (All)



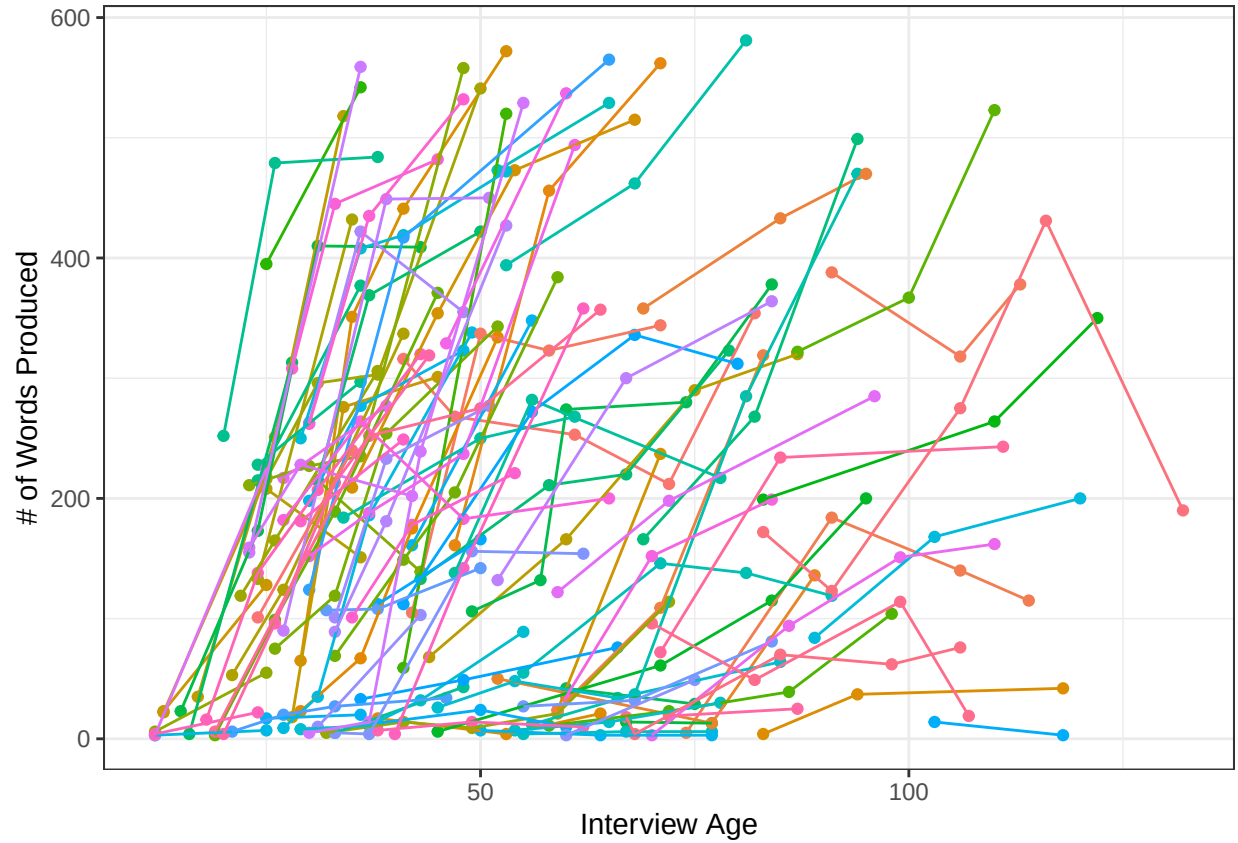
111 with between 9 and 15 months distance between CDIs

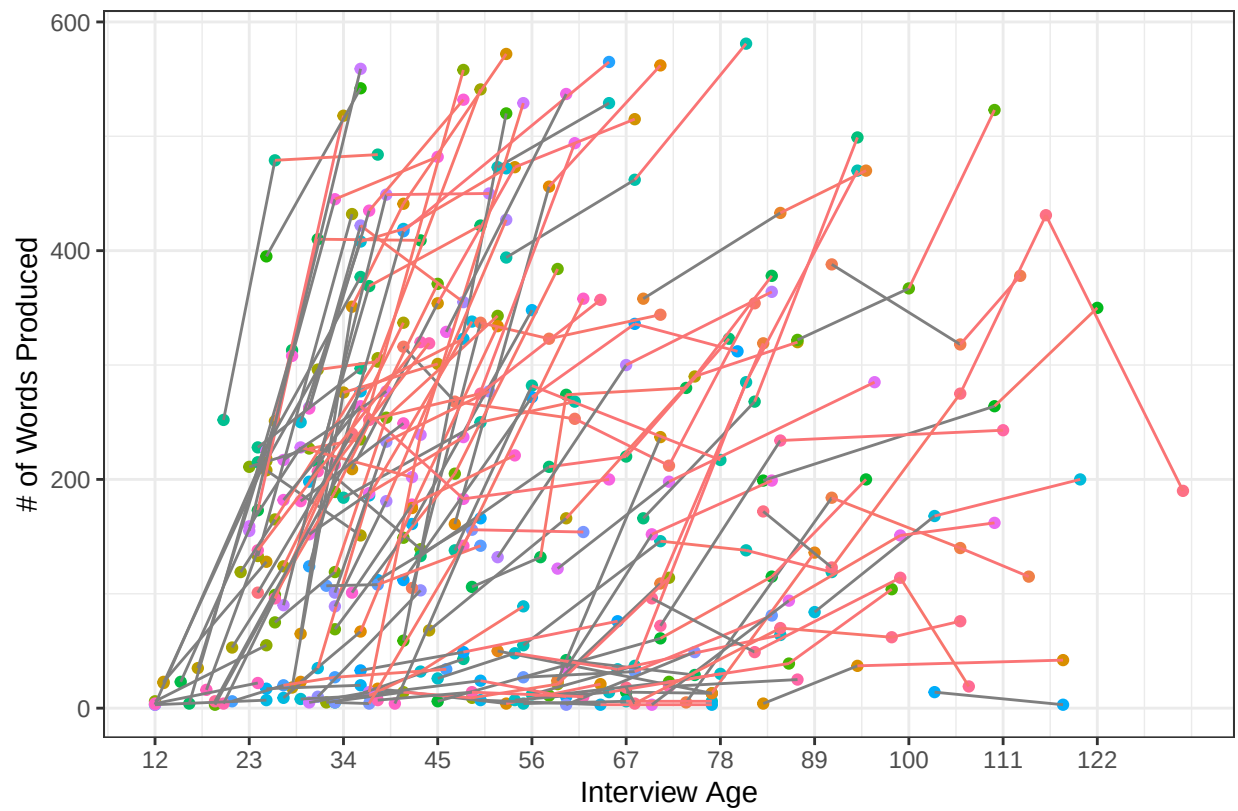
Distributions of first interviews



Spaghetti Plots

Plots showing the progression of productive vocabulary over interview age. The majority of kids show increases in productive vocabulary over time, but this is not always the case. The next page digs deeper into those cases.

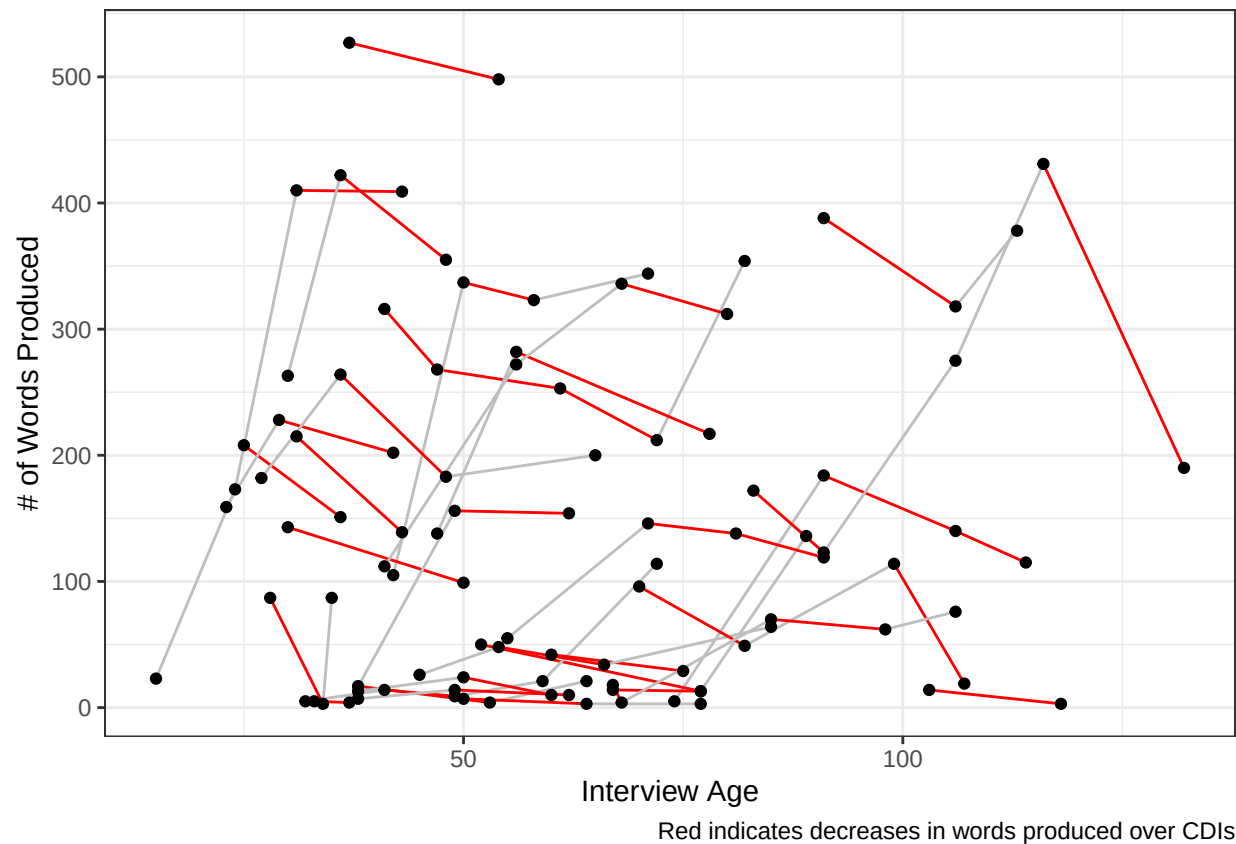




Red connections are 9–15 month gaps between CDIs

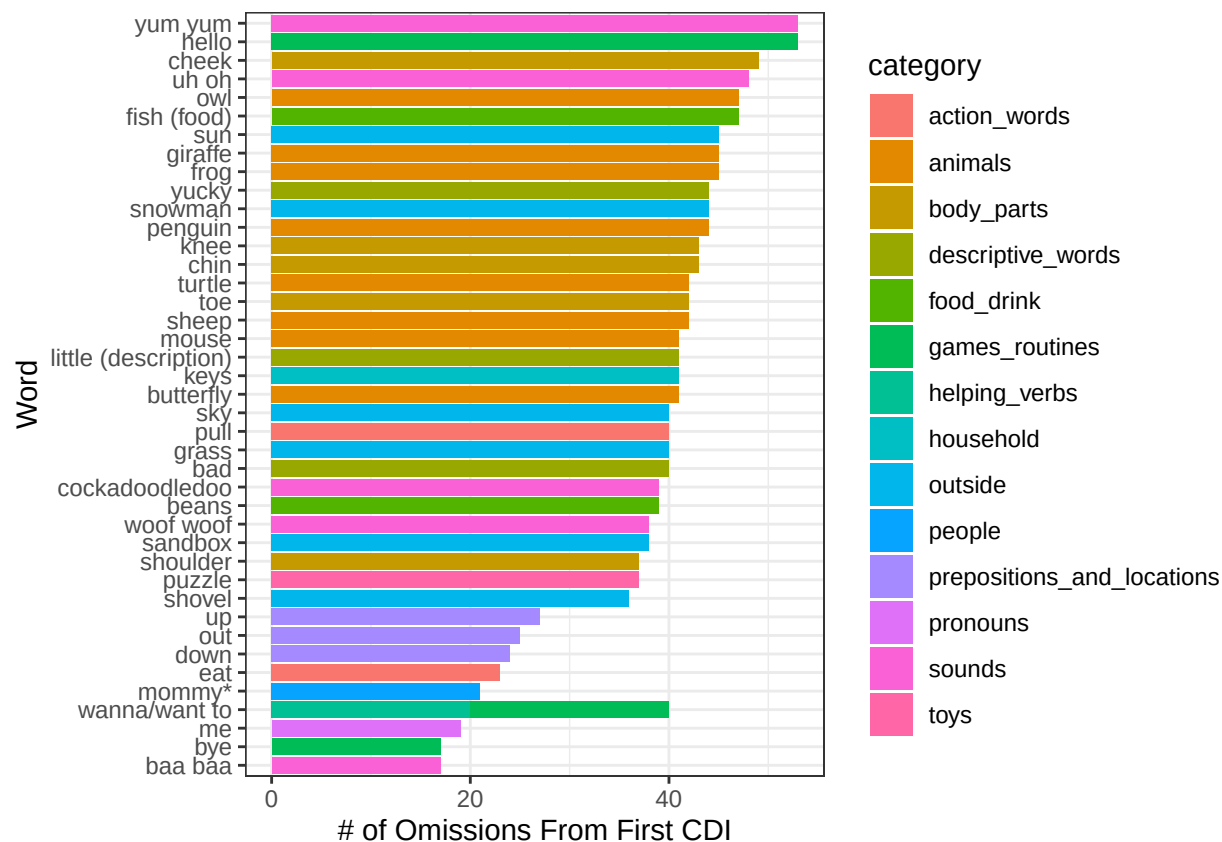
Inconsistent gains in vocabulary over time

Some of the kids show drops in vocabulary size between CDI administrations. There are 30 kids who have longitudinal CDIs that decrease at some point in development. These may be considerations for removal as they do not follow the pattern of development of the majority of the kids. It is impossible to know if this is due to bad reporting or legitimate losses in productive vocabulary.



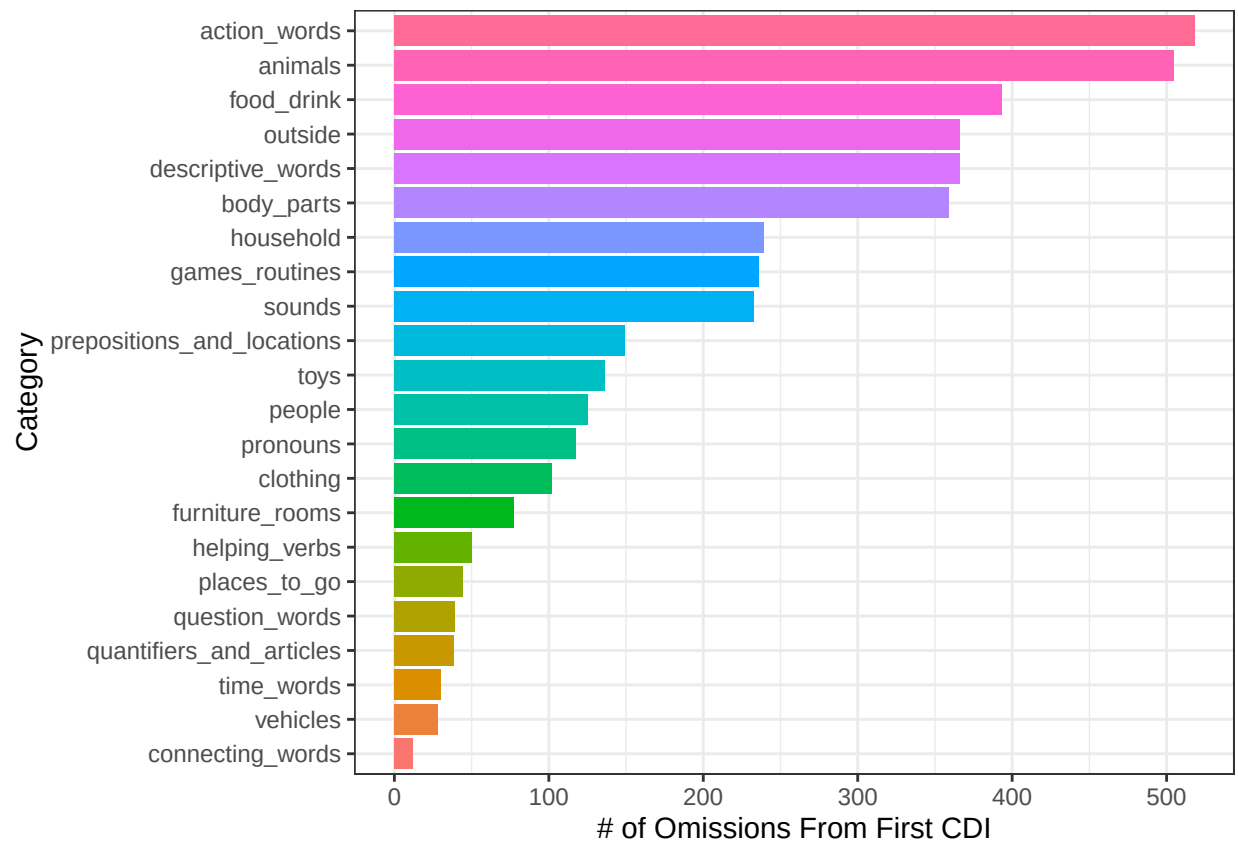
Mean Vocabulary Decreases	SD	Range
-37.13158	43.35409	-241
-37.13158	43.35409	-1

Top 40 words not consistently reported across CDIs.



Here, we are looking at item-level distributions of the top 50 words that were reported as being known in the first CDI administration, but not reported as being known in at least one of the subsequent administrations. The bars are filled by CDI category. Values on the x-axis report the number of times the word was reported on the first administration of the CDI but not later CDIs.

Overall distribution of omissions across categories



The majority of words not being included in subsequent CDIs are action words, followed by animals and food and drinks.